

Online Video Meeting System: Architectural & Technical Guide

End-to-End WebRTC Signaling, Dual-Pipeline Audio Decoding, Voice Activity Detection, and Auto-Archiving Suite

1. System Architecture & High-Level Overview

The NITTE Online Video Meeting platform provides an encrypted, browser-to-browser counseling environment designed for students and Relationship Officers (ROs). Built on modern **WebRTC standards**, the system establishes direct peer-to-peer audio/video transmission with automated serverless cloud archiving directly attached to student support tickets.

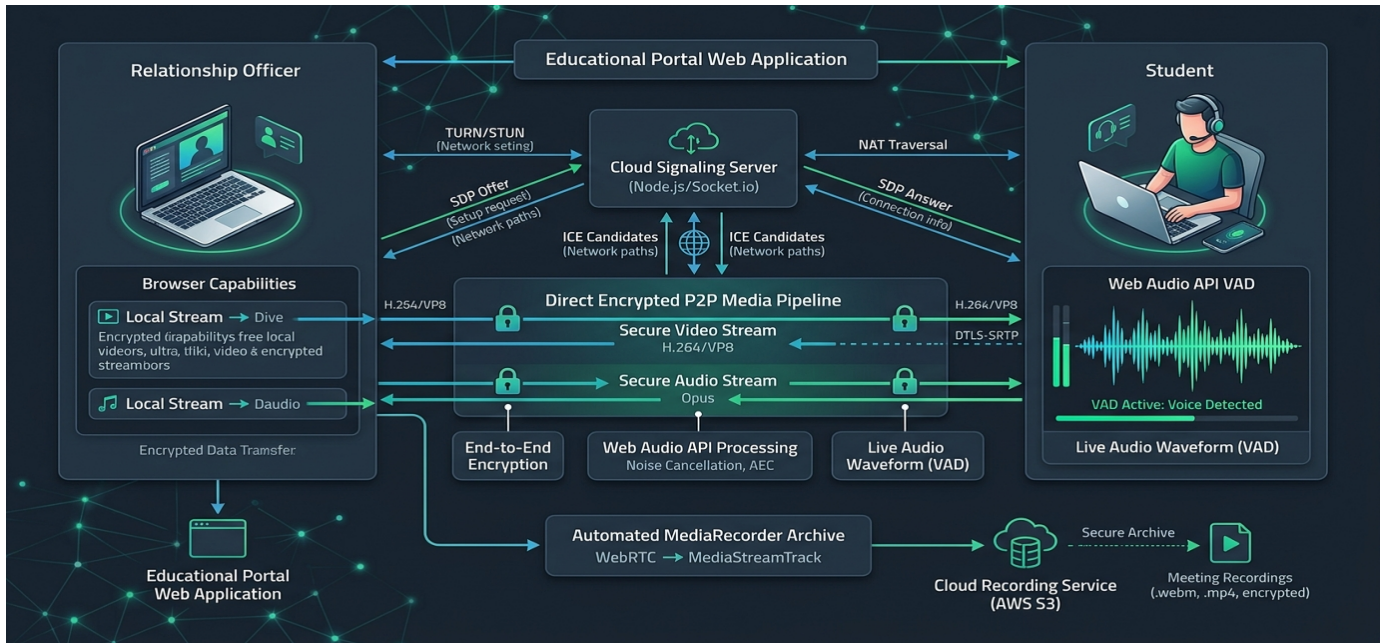


Figure 1: Full-Duplex WebRTC P2P Signaling Pipeline, Audio/Video Streams, VAD Energy Detection, and MediaRecorder Archive

Key Capabilities of the WebRTC Pipeline:

- **Dual-Channel Signaling Delivery:** Employs an ultra-fast cross-tab BroadcastChannel('nitte_webrtc_signals') for 0ms same-device coordination and resilient backend HTTP polling via /api/meetings/signals/:issued for WAN traversal.
- **Session Instance Tracking (Anti-Desync):** Every active connection is stamped with a unique sessionId. When a participant leaves or rejoins, stale RTCPeerConnection instances are torn down and re-initialized in under 200ms without page reloads.
- **Opus In-Band FEC & Bitrate Clamping:** SDP tuning injects useinbandfec=1 with minptime=10, reconstructing dropped voice packets over congested Wi-Fi while clamping video at 900 kbps (30fps fluid motion).

Pipeline Component	Configuration / Specification	Engineered Advantage
Video Resolution	640 x 480 @ 30fps (Standard HD)	Prevents USB camera driver exhaustion during dual-tab calls
Audio Processing	AEC, Noise Suppression, AGC	Eliminates speaker acoustic feedback and ambient room noise
Video Bitrate	900 kbps (maintain-framerate)	Guarantees silky 30fps motion without bufferbloat
Audio Codec	Opus 48 kbps, FEC enabled	Zero round-trip latency packet reconstruction over Wi-Fi
Encryption	DTLS-SRTP 256-bit AES	End-to-end cryptographic privacy (No intermediary eavesdropping)
Recording Format	video/webm (MediaRecorder)	Relaxed 4s buffer chunking; auto-attaches to ticket history

2. Modern 1120px Conference Room User Experience

The conference stage has been revamped to Google Meet and Zoom caliber aesthetics, featuring a wide 1120px dark obsidian stage (#0b1120), dynamic 16:9 HD video cards, and a floating frosted-glass controls dock with blur glassmorphism.

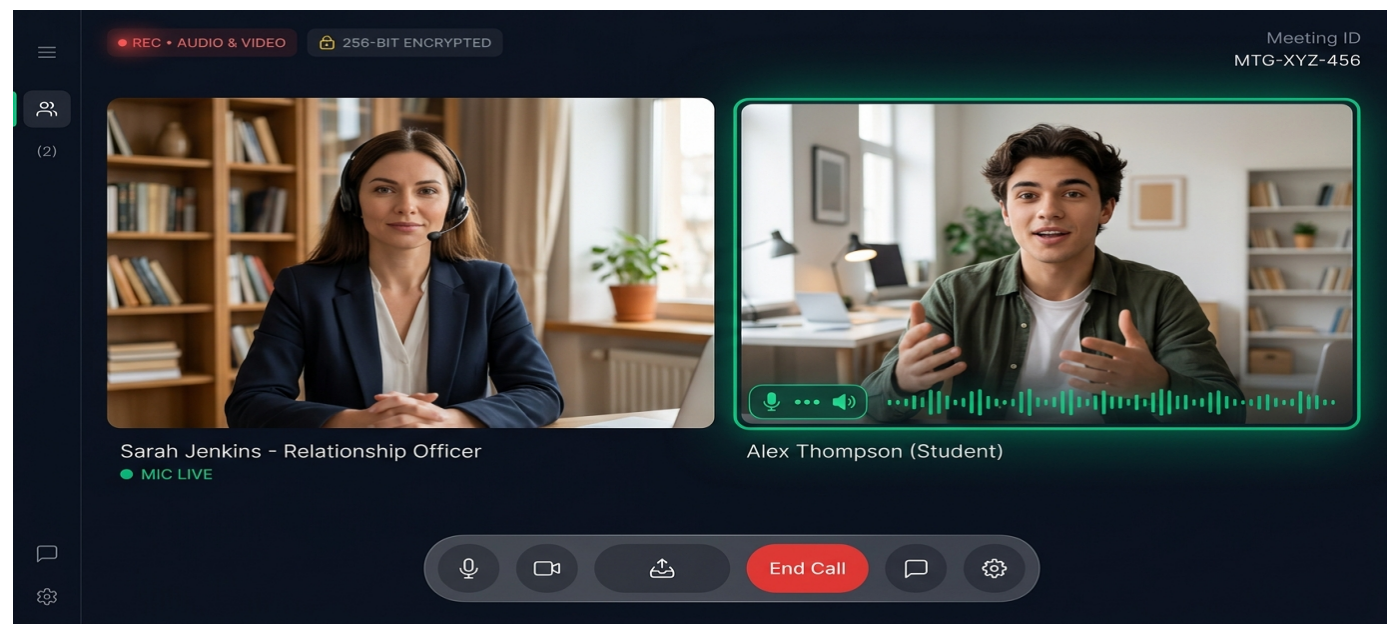


Figure 2: Modern 1120px Conference Stage with Active-Speaker Emerald Halo, Live Audio Waveform Equalizer, and Frosted-Glass Controls Dock

3. Dual-Pipeline Audio Architecture & Voice Activity Detection (VAD)

A critical engineering challenge in browser-based WebRTC is microphone silence caused by Chromium audio track multiplexing locks and aggressive autoplay restrictions. The NITTE meeting engine employs a **Dual-Pipeline Audio Architecture** to solve this completely:

- **Dedicated <audio> Decoding Element:** An independent <audio ref={audioRef} autoPlay playsInline /> element is bound directly to incoming remote streams, bypassing video decoding pauses.
- **Dynamic addtrack Event Binding:** Automatically hooks newly negotiated audio tracks even if they arrive seconds after video playback initiates.
- **Web Audio API AnalyserNode:** Calculates root-mean-square (RMS) energy levels every 100ms. If volume exceeds 8%, the participant's video card triggers a radiant emerald halo (boxShadow: 0 0 0 2px #10b981) alongside an animated soundwave equalizer.
- **Autoplay-Restriction Banner:** If browser security holds unmuted media, a floating 'Click to Unmute Audio' prompt allows instant recovery.

University Compliance & Cloud Archiving Guarantee

Every online video session between a student and Relationship Officer is treated as an official counseling record. Audio and video feeds are recorded client-side by the host RO via MediaRecorder, compiled into standard .webm blobs, and permanently archived to the student's ticket history upon call termination for auditing, playback, and offline download.

4. End-to-End Meeting Lifecycle Summary

- **Step 1: Scheduling:** RO schedules an 'Online Video Meeting' in the Schedule Modal. An encrypted room session is generated, and student is notified with date and time.
- **Step 2: Initiation:** RO launches the meeting; local camera and microphone stream initialize at 640x480. MediaRecorder begins buffering audio/video chunks.
- **Step 3: Student Join:** Student clicks 'Join Live Session' on their ticket. WebRTC signals exchange SDP offer/answer with Opus FEC enabled.
- **Step 4: Active Stage:** Encrypted peer-to-peer call active. Live Voice Activity Detection displays active-speaker halos and soundwave equalizers on speaking party.
- **Step 5: Termination:** RO clicks 'Stop Recording & End Meeting'. Session sends termination broadcast, auto-compiles .webm video, and archives recording to ticket.